The three wave functions u (exact), $\tilde{u}$, and $\bar{u}$ are shown in Fig. 42 i correct normalization (which, for u, can only be found by numerica computation). The two approximations are too large at small value of r which, however, have rather a small effect upon both, the normali zation and the energy integral, since the volume element reduces thei weight by a factor r^2. This deviation is compensated by values toc small at larger r. The asymptotic behaviour shows no large differences between the three curves:

$$u a^{\frac{3}{2}} \simeq \bar{u} a^{\frac{3}{2}} \to 0.308\,\mathrm{e}^{-x}/x; \qquad \tilde{u} a^{\frac{3}{2}} \to 0.346\,\mathrm{e}^{-x}/x$$

with $x = r/(2a)$.

Problem 76. Momentum space wave functions for central force potentials

To show that the splitting off of a spherical harmonic in the wave function in ordinary space permits the same factorization in momentum space.

Solution. In the Fourier transform

$$u(\boldsymbol{r}) = \frac{1}{(2\pi)^{\frac{3}{2}}} \int d^3k\, \mathrm{e}^{i\boldsymbol{k}\cdot\boldsymbol{r}} f(\boldsymbol{k}), \tag{76.1}$$

$$f(\boldsymbol{k}) = \frac{1}{(2\pi)^{\frac{3}{2}}} \int d^3x\, \mathrm{e}^{-i\boldsymbol{k}\cdot\boldsymbol{r}} u(\boldsymbol{r}) \tag{76.2}$$

it shall be supposed that u is factorized:

$$u(\boldsymbol{r}) = \frac{1}{r} \chi_l(r)\, Y_{l,m}(\vartheta, \varphi). \tag{76.3}$$

In order to find its Fourier transform according to (76.2), let us expand the exponential into spherical harmonics of the angle γ between the vectors $\boldsymbol{r}$ in the direction ϑ, φ and $\boldsymbol{k}$ in the direction Θ, Φ (cf. Problem 81):

$$\mathrm{e}^{-i\boldsymbol{k}\cdot\boldsymbol{r}} = \sum_{\lambda=0}^{\infty} \sqrt{4\pi(2\lambda+1)}\, i^{-\lambda} \frac{j_\lambda(kr)}{kr} Y^*_{\lambda,0}(\cos\gamma). \tag{76.4}$$

Here $Y_{\lambda,0}(\cos\gamma)$ may be expressed by the polar angles of $\boldsymbol{r}$ and $\boldsymbol{k}$ using the addition theorem of spherical harmonics,

$$Y_{\lambda,0}(\cos\gamma) = \sqrt{\frac{4\pi}{2\lambda+1}} \sum_{\mu=-\lambda}^{+\lambda} Y^*_{\lambda,\mu}(\Theta,\Phi)\, Y_{\lambda,\mu}(\vartheta,\varphi). \tag{76.5}$$

Setting (76.5) into (76.4), and the result and (76.3) into (76.2), we get

$$f(\mathbf{k}) = \frac{4\pi}{(2\pi)^{\frac{3}{2}}} \int_0^\infty dr\, r^2 \oint d\Omega_r \sum_{\lambda\mu} i^{-\lambda} \frac{j_\lambda(kr)}{kr} Y_{\lambda,\mu}(\Theta,\Phi)\, Y^*_{\lambda,\mu}(\vartheta,\varphi) \frac{\chi_l}{r} Y_{l,m}(\vartheta,\varphi).$$

The integration over all directions of the vector $\boldsymbol{r}$ can be performed:

$$\oint d\Omega_r\, Y^*_{\lambda,\mu}(\vartheta,\varphi)\, Y_{l,m}(\vartheta,\varphi) = \delta_{l\lambda}\,\delta_{m\mu}$$

so that only one term (l,m) remains of the double sum:

$$k f(\mathbf{k}) = \sqrt{\frac{2}{\pi}}\, i^{-l} \int_0^\infty dr\, j_l(kr)\, \chi_l(r) \cdot Y_{l,m}(\Theta,\Phi),$$

i.e. the momentum space function, $f(\mathbf{k})$, may be factorized in the form

$$f(\mathbf{k}) = \frac{1}{k} g_l(k)\, Y_{l,m}(\Theta,\Phi) \tag{76.6}$$

with

$$g_l(k) = \sqrt{\frac{2}{\pi}}\, i^{-l} \int_0^\infty dr\, j_l(kr)\, \chi_l(r). \tag{76.7}$$

The radial parts $g_l(k)$ and $i^{-l}\chi_l(r)$ therefore stand in the mutual relation of a Hankel integral transform, the inversion of (76.7) being

$$\chi_l(r) = \sqrt{\frac{2}{\pi}}\, i^{l} \int_0^\infty dk\, j_l(kr)\, g_l(k). \tag{76.8}$$

If we normalize the space function u in the usual way to permit its probability interpretation,

$$\int_0^\infty dr\, |\chi_l(r)|^2 = 1, \tag{76.9}$$

we find by putting (76.8) into (76.9):

$$\frac{2}{\pi} \int_0^\infty dr \int_0^\infty dk \int_0^\infty dk'\, j_l(kr)\, g_l(k)\, j_l(k'r)\, g_l^*(k') = 1.$$

Performing first the integration over r,

$$\int_0^\infty dr j_l(kr) j_l(k'r) = \frac{\pi}{2}\delta(k-k'), \tag{76.10}$$

we get

$$\int_0^\infty dk |g_l(k)|^2 = 1, \tag{76.11}$$

i.e. the same probability interpretation holds in momentum space: in the quantum state under consideration the particle will be found with the absolute value of momentum between k and $k+dk$ with a probability $|g_l(k)|^2\,dk$.

Problem 77. Momentum space integral equation for central force potentials

In Problem 14 a general integral equation has been established for the momentum space wave functions. It shall be shown that for a central force field the solutions can be factorized in the form

$$f(\boldsymbol{k}) = \frac{1}{k} g_l(k)\, Y_{l,m}(\Theta, \Phi). \tag{77.1}$$

The special integral equation for $g_l(k)$ shall then be derived for the hydrogen atom.

Solution. The integral equation (14.6), written in atomic units,

$$\left(\tfrac{1}{2}k^2 - E\right) f(\boldsymbol{k}) = -\int d^3k'\, W(\boldsymbol{k}-\boldsymbol{k}')\, f(\boldsymbol{k}') \tag{77.2}$$

with

$$W(\boldsymbol{k}) = \frac{1}{8\pi^3}\int d^3x\, \mathrm{e}^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\, V(\boldsymbol{r}) \tag{77.3}$$

shall be reduced to a radial equation for $g_l(k)$ if $V(\boldsymbol{r})$ depends on the absolute value r only of the vector $\boldsymbol{r}$. In that case, (77.3) may be integrated over the solid angle, the result depending only upon the absolute value of the vector $\boldsymbol{k}$:

$$W(k) = \frac{4\pi}{8\pi^3}\int_0^\infty dr\, r^2\, V(r) \frac{\sin kr}{kr}. \tag{77.4}$$

The kernel of the integral equation then becomes a function of

$$(\boldsymbol{k}-\boldsymbol{k}')^2 = k^2 + k'^2 - 2kk'\cos\gamma \tag{77.5}$$

where γ is the angle between the vectors $\boldsymbol{k}$ and $\boldsymbol{k}'$. The function W may then be expanded into a series of Legendre polynomials,

$$W(|\boldsymbol{k}-\boldsymbol{k}'|) = \sum_{n=0}^{\infty} a_n(k,k') P_n(\cos\gamma) \tag{77.6}$$

whose coefficients a_n depend on the absolute values k and k' only. This is the essential point leading to possible factorization.

Using (77.1) for $f(\boldsymbol{k})$, the integral equation (77.2) now becomes

$$(\tfrac{1}{2}k^2 - E)\frac{1}{k} g_l(k)\, Y_{l,m}(\Theta,\Phi)$$

$$= -\sum_n \int_0^\infty dk'\, k'^2 a_n(k,k') \frac{1}{k'} g_l(k') \oint d\Omega'\, P_n(\cos\gamma)\, Y_{l,m}(\Theta',\Phi').$$

The angular integral can be evaluated using the addition theorem

$$P_n(\cos\gamma) = \frac{4\pi}{2n+1} \sum_{\mu=-n}^{n} Y^*_{n,\mu}(\Theta',\Phi')\, Y_{n,\mu}(\Theta,\Phi)$$

which reduces the sums to one term only with $n=l$, $\mu=m$:

$$\frac{1}{k} g_l(k)\, Y_{l,m}(\Theta,\Phi) = -\frac{4\pi}{2l+1} \int_0^\infty dk'\, k'\, a_l(k,k') g_l(k')\, Y_{l,m}(\Theta,\Phi).$$

This is an identity in the polar angles thus showing that the factorization (77.1) is correct, and leaving us with the radial integral equation

$$(\tfrac{1}{2}k^2 - E) g_l(k) = -\frac{4\pi k}{2l+1} \int_0^\infty dk'\, k'\, a_l(k,k') g_l(k'). \tag{77.7}$$

If the central force is the *Coulomb attraction* of the hydrogen atom,

$$V(r) = -\frac{1}{r}, \tag{77.8}$$

the integral (77.4) can be solved using the limiting relation[17]

$$\lim_{\varepsilon\to 0} \int_0^\infty dx\, e^{-\varepsilon x} \sin x = 1$$

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[17] The Fourier transform (77.3) of the potential is essentially the Born amplitude the convergence of which is limited by Coulomb behaviour at large values of r. Cf. Problem 105.

so that we find

$$W(k) = -\frac{1}{2\pi^2 k^2}, \tag{77.9}$$

and, according to (77.5),

$$W(\boldsymbol{k}-\boldsymbol{k}') = -\frac{1}{2\pi^2|\boldsymbol{k}-\boldsymbol{k}'|^2} = -\frac{1}{4\pi^2 k k'(z-\cos\gamma)}$$

with

$$z = \frac{k^2+k'^2}{2kk'}. \tag{77.10}$$

Putting $\cos\gamma = t$, Eq. (77.6) then is the well-known expansion of

$$\frac{1}{z-t} = \sum_{n=0}^{\infty}(2n+1)Q_n(z)P_n(t) \tag{77.11}$$

where the coefficients

$$Q_n(z) = \frac{1}{2}\int_{-1}^{+1}\frac{dt\,P_n(t)}{z-t} \tag{77.12}$$

are the Legendre functions of the second kind. We therefore find

$$a_n(k,k') = -\frac{1}{4\pi^2 kk'}(2n+1)Q_n\left(\frac{k^2+k'^2}{2kk'}\right)P_n(\cos\gamma), \tag{77.13}$$

so that finally we arrive at the radial integral equation

$$(\tfrac{1}{2}k^2 - E)g_l(k) = \frac{1}{\pi}\int_0^\infty dk'\,Q\left(\frac{k^2+k'^2}{2kk'}\right)g_l(k'). \tag{77.14}$$

Literature. The integral equation (77.14) was solved by Fock, V.: Z. Physik 98, 145 (1935). This way means determination of the momentum space eigenfunctions without any recourse to coordinate space functions. The latter way, however, proves simpler for the Coulomb field and shall be given in the following problem.

Problem 78. Momentum space wave functions for hydrogen

To determine the momentum space wave functions for the lowest levels ($1s, 2s, 2p$) of the hydrogen atom.

Solution. Since in Problem 76 we have shown that, in a central field, a factorized coordinate wave function

$$u(\boldsymbol{r}) = \frac{1}{r}\chi_l(r)\,Y_{l,m}(\vartheta,\varphi) \tag{78.1}$$

leads to a factorized momentum space wave function

$$f(\mathbf{k}) = \frac{1}{k} g_l(k)\, Y_{l,m}(\Theta, \Phi), \tag{78.2}$$

we have only to determine the radial part $g_l(k)$ which follows from a Hankel transform of $\chi_l(r)$:

$$g_l(k) = \sqrt{\frac{2}{\pi}}\, i^{-l} \int_0^\infty dr\, j_l(kr)\, \chi_l(r). \tag{78.3}$$

For the three chosen states we have, in atomic units,

1s: $\chi_{10}(r) = 2r\mathrm{e}^{-r}$,

$$g_{10} = \sqrt{\frac{2}{\pi}}\, 2 \int_0^\infty dr\, r \sin kr\, \mathrm{e}^{-r};$$

2s: $\chi_{20}(r) = \frac{1}{\sqrt{2}}(r - \frac{1}{2}r^2)\mathrm{e}^{-\frac{r}{2}}$,

$$g_{20} = \frac{1}{\sqrt{\pi}} \int_0^\infty dr (r - \tfrac{1}{2}r^2) \sin kr\, \mathrm{e}^{-\frac{r}{2}};$$

2p: $\chi_{21}(r) = \frac{1}{\sqrt{24}} r^2 \mathrm{e}^{-\frac{r}{2}}$,

$$g_{21} = -i \frac{1}{\sqrt{12\pi}} \int_0^\infty dr\, r^2 \left(\frac{\sin kr}{kr} - \cos kr\right) \mathrm{e}^{-\frac{r}{2}}.$$

Evaluation of these integrals is elementary, though a little cumbersome. The results are

$$g_{10}(k) = \sqrt{\frac{2}{\pi}} \frac{4k}{(1+k^2)^2}; \tag{78.4a}$$

$$g_{20}(k) = \frac{32}{\sqrt{\pi}} \frac{k(1-4k^2)}{(1+4k^2)^3}; \tag{78.4b}$$

$$g_{21}(k) = -i \frac{128}{\sqrt{3\pi}} \frac{k^2}{(1+4k^2)^3}. \tag{78.4c}$$

In all cases there holds the normalization law

$$\int_0^\infty dk\,|g_{nl}(k)|^2 = 1 \tag{78.5}$$

which may be checked directly for each of the functions (78.4a–c), but which also follows from the general theory, cf. (76.11).

Problem 79. Stark effect of a three-dimensional rotator

To calculate the Stark effect in second approximation of perturbation for a three-dimensional free rotator with the electric dipole moment p.

Solution. The unperturbed Schrödinger equation is solved by spherical harmonics:

$$\frac{\hbar^2}{2\Theta} L^2\, Y_{l,m} = E_l\, Y_{l,m} \tag{79.1}$$

with eigenvalues

$$E_l = \frac{\hbar^2\, l(l+1)}{2\Theta}. \tag{79.2}$$

The perturbation energy for an electrical field $\mathscr{E}$ applied in arbitrary direction is

$$V = -p(\mathscr{E}_x \sin\vartheta\cos\varphi + \mathscr{E}_y \sin\vartheta\sin\varphi + \mathscr{E}_z\cos\vartheta). \tag{79.3}$$

To find the matrix elements of V,

$$\langle l', m'|V|l, m\rangle = \oint d\Omega\, Y^*_{l',m'}\, V(\vartheta,\varphi)\, Y_{l,m}, \tag{79.4}$$

we use the relations[18]

$$\begin{aligned} \sin\vartheta\, e^{i\varphi}\, Y_{l,m} &= a_{l,m}\, Y_{l+1,m+1} - a_{l-1,-m-1}\, Y_{l-1,m+1},\\ \sin\vartheta\, e^{-i\varphi}\, Y_{l,m} &= -a_{l,-m}\, Y_{l+1,m-1} + a_{l-1,m-1}\, Y_{l-1,m-1},\\ \cos\vartheta\, Y_{l,m} &= b_{l,m}\, Y_{l+1,m} + b_{l-1,m}\, Y_{l-1,m} \end{aligned} \tag{79.5}$$

with the abbreviations

$$a_{l,m} = \sqrt{\frac{(l+m+1)(l+m+2)}{(2l+1)(2l+3)}}; \qquad b_{l,m} = \sqrt{\frac{(l+m+1)(l-m+1)}{(2l+1)(2l+3)}}. \tag{79.6}$$

[18] Cf. e.g. Bethe, H.A., Salpeter, E.E., p. 432, in: *Encyclopedia of Physics*, vol. 35. Berlin-Göttingen-Heidelberg: Springer 1957.